

HARSH

Senior Mobile & Full Stack Engineer

Results-driven Senior Engineer with **9+ years** of hands-on experience architecting and delivering enterprise-grade mobile and full-stack applications. Specialized in cross-platform mobile development using **Xamarin.Forms, .NET MAUI, and Flutter**, coupled with robust backend systems built on **ASP.NET Core, FastAPI, SQL Server, and Supabase**. Passionate about integrating **AI/ML capabilities** into production applications using Python, Ollama, and modern LLM frameworks. Proven track record across healthcare, hospitality, logistics, finance, and real-time communication domains.

9+

Years Experience

11+

Projects Delivered

5

Platforms

3+

AI Projects

CAREER TRAJECTORY

■ Senior Software Engineer

Jan 2026 – Present

Leading architecture & development of enterprise mobile apps, AI integration, and full-stack solutions.

■ Xamarin Developer

Jan 2022 – Dec 2025

Designed and delivered cross-platform mobile applications for healthcare, logistics, and hospitality using Xamarin.Forms & .NET MAUI.

■ Associate Software Engineer

Jun 2017 – Dec 2021

Built foundational mobile and backend systems, REST APIs, database design, and client-facing applications.

CORE SKILLS & TECHNOLOGIES

MOBILE	.NET MAUI · Xamarin.Forms · Flutter · Dart · C# · MVVM
BACKEND	ASP.NET Core · FastAPI · REST APIs · Entity Framework · Python
AI / ML	Ollama · LLM Integration · AI Pipelines · Computer Vision
CLOUD & DB	SQL Server · Supabase · SQLite · Azure · Serverless
TOOLS	Visual Studio · VS Code · Git · Postman · Agile / Scrum
OTHER	HIPAA · HL7 · Real-time Chat · BLE · Cross-platform

PROJECT PORTFOLIO (11 Projects)

A curated collection of production applications spanning healthcare, AI, hospitality, logistics, and finance — demonstrating depth in mobile engineering, backend architecture, and AI integration.

1 HealthxApp for Patients

Enterprise Healthcare Platform – Patient Portal

A comprehensive patient-facing mobile application enabling appointment scheduling, secure document management, HIPAA-compliant video consultations, and real-time chat with healthcare providers.

- Built end-to-end appointment booking with provider availability and calendar sync
- Implemented secure video consultation using WebRTC with encrypted streams
- Developed real-time chat system with message history and push notifications
- Integrated HL7/FHIR standards for healthcare data interoperability
- Document upload/download with secure cloud storage and access control
- HIPAA-compliant data handling with encryption at rest and in transit
- Offline-first architecture with background sync for poor connectivity scenarios
- Role-based access control separating patient, provider, and admin flows

Tech: [Xamarin.Forms] · [C#] · [ASP.NET Core API] · [SQL Server] · [HL7] · [HIPAA] · [MVVM] · [Push Notifications]

2 HealthxApp for Providers

Enterprise Healthcare Platform – Provider Dashboard

Provider-facing companion application enabling doctors and clinical staff to manage appointments, review patient documents, conduct video consultations, and communicate securely with patients.

- Appointment management dashboard with schedule overview and conflict detection
- Patient document review system with annotation and approval workflows
- Video consultation initiation with recording capability for medical records
- Secure messaging with read receipts, typing indicators, and file sharing
- Integration with hospital EMR systems via HL7 interfaces
- Multi-location support for providers practising across clinics
- Analytics dashboard showing consultation metrics and patient engagement
- Notification system for urgent patient requests and schedule changes

Tech: [Xamarin.Forms] · [C#] · [ASP.NET Core API] · [SQL Server] · [HL7] · [HIPAA] · [MVVM] · [WebRTC]

3 Clipboard

Patient Intake & Clinical Workflow Management

Digital patient intake solution replacing paper-based processes — handling appointments, consent forms, questionnaires, and medical history management with seamless offline support.

- Dynamic consent form builder with e-signature capture and PDF generation
- Configurable medical questionnaires with conditional logic and branching
- Complete medical history intake with allergies, medications, and conditions
- Offline-first SQLite storage with intelligent background synchronisation
- Migrated from Xamarin.Forms to .NET MAUI for long-term platform support
- Barcode scanning for patient identification and record lookup
- Multi-language support for diverse patient demographics
- Seamless integration with backend .NET Core APIs and SQL Server

Tech: [Xamarin.Forms] · [.NET MAUI] · [C#] · [SQLite] · [SQL Server] · [.NET Core API] · [MVVM] · [Offline Sync]

4 SMD Mobile

Hospital & Clinic Management System

Comprehensive hospital and clinic management mobile application covering patient records, medicine inventory, appointment scheduling, and clinical workflow automation.

- Patient registration and record management with search and filtering
- Medicine inventory tracking with stock alerts and expiry management
- Appointment scheduling with provider calendars and automated reminders
- Migrated entire codebase from Xamarin.Forms to .NET MAUI
- Improved startup time and memory efficiency via framework optimisation
- REST API integration for secure, real-time backend communication
- Resolved Android/iOS platform-specific UI and compatibility issues
- Extensive testing, debugging, and production performance tuning

Tech: [Xamarin.Forms] · [.NET MAUI] · [C#] · [SQLite] · [SQL Server] · [.NET Core API] · [MVVM] · [REST APIs]

5 LogicWash

Commercial Car Wash Management Solution

A streamlined mobile solution for managing commercial car wash operations — handling service queues, customer management, wash tracking, and operational analytics.

- Service queue management with real-time status tracking
- Customer profile management with wash history and preferences
- Multiple wash package configuration with pricing tiers
- Offline-capable SQLite storage for uninterrupted operations
- Staff scheduling and performance tracking dashboards
- Revenue reporting with daily, weekly, and monthly breakdowns
- Push notifications for service completion and promotional offers
- Clean MVVM architecture ensuring maintainability and testability

Tech: [.NET MAUI] · [C#] · [SQLite] · [MVVM] · [Local Storage] · [Cross-platform]

6 Digital Butler

AI-Powered Hotel & Resort Guest Management Platform

An intelligent hospitality platform leveraging AI to provide personalised guest experiences — from smart room booking and concierge services to AI-driven recommendations and guest communication.

- AI-powered concierge chatbot using Ollama LLM for natural guest interactions
- Smart room booking with dynamic pricing and availability optimisation
- Guest preference learning system for personalised service recommendations
- Real-time communication between guests and hotel staff via chat
- Integration with hotel PMS for room status, housekeeping, and maintenance
- Multi-property support with centralised management dashboard
- Supabase real-time database for instant updates across all clients
- Python FastAPI backend with AI pipeline orchestration

Tech: [Flutter] · [Dart] · [Supabase] · [Python] · [FastAPI] · [AI] · [Ollama] · [Real-time DB] · [Cross-platform]

7 SmartLogix

AI-Powered Shipment & Logistics Tracking Platform

Intelligent logistics platform that uses AI for shipment registration, route optimisation, delivery prediction, and real-time tracking — transforming traditional shipping workflows.

- AI-driven shipment registration with automatic data extraction from documents
- Real-time GPS tracking with estimated delivery time predictions
- Route optimisation engine using ML models for cost and time efficiency
- Automated status updates with intelligent notification triggers
- Document scanning and OCR for bills of lading and shipping labels
- Analytics dashboard with shipment performance metrics and trends
- Multi-carrier integration with unified tracking interface
- Supabase real-time sync for instant status propagation

8 ReconVision

AI Platform for Crime & Anomaly Detection

Advanced AI-powered surveillance and analytics platform combining computer vision for crime detection, facial recognition, violence detection, and behavioural analytics in real-time video feeds.

- Real-time video feed analysis with computer vision models for anomaly detection
- Face detection and recognition with configurable alert thresholds
- Violence and aggressive behaviour detection using trained ML models
- Analytics dashboard with heat maps, incident timelines, and trend analysis
- Multi-camera support with centralised monitoring interface
- Alert system with configurable escalation workflows
- Historical incident search with video playback and evidence export
- Python AI pipeline with Ollama integration for contextual analysis

Tech: [Flutter] · [Dart] · [Supabase] · [Python] · [FastAPI] · [AI] · [Ollama] · [Computer Vision] · [Analytics]

9 Pinglet

Real-Time Chat Application

A modern, performant real-time messaging application with instant message delivery, presence indicators, media sharing, and group conversation support.

- Real-time messaging with sub-second delivery using Supabase Realtime
- Online/offline presence indicators with last-seen timestamps
- Media sharing including images, documents, and voice messages
- Group chat creation and management with admin controls
- Message read receipts and typing indicators
- Push notifications for background message delivery
- End-to-end message history with search functionality
- Clean Material Design UI with smooth animations

Tech: [Flutter] · [Dart] · [Supabase] · [Realtime] · [Push Notifications] · [Cross-platform]

10 Finance App

Personal Finance & Expense Tracking

A personal finance management application enabling users to track expenses, set budgets, visualise spending patterns, and gain insights into their financial habits.

- Expense recording with category tagging and receipt photo capture
- Budget creation with configurable limits and overspend alerts
- Interactive charts and graphs for spending pattern visualisation
- Monthly and yearly financial summaries with export capability
- Recurring expense tracking with automatic categorisation
- Multi-currency support with real-time conversion rates
- Secure authentication with biometric login support
- Cloud sync via Supabase for multi-device access

Tech: [Flutter] · [Dart] · [Supabase] · [Charts] · [Cross-platform] · [Biometric Auth]

11 GalleryHub

Cross-Platform Smart Gallery & Media Manager

A cross-platform gallery application built with Flutter — ported from .NET MAUI. Features rich media browsing with grid/list layouts, folder organisation, full-resolution image viewing, native video playback, camera capture, and batch operations.

- Gallery view with grid (2/3/4 columns) or list layout, grouped by date
- Folder browsing with media organized by albums with cover thumbnails
- Full-resolution image viewer with pinch-to-zoom using PhotoView
- Native video player with play/pause and progress scrubbing
- Share single or multiple media items via system share sheet

- Delete and rename media files with platform-specific handling
- Camera capture for photos and videos directly from the app
- Theme support with System, Light, and Dark modes with persistent preference
- Multi-select selection mode for batch delete or share operations
- Animated auto-hiding header and floating pill-shaped bottom navigation

Tech: [Flutter] · [Dart] · [Provider] · [photo_manager] · [video_player] · [Cross-platform]

This portfolio represents a curated selection of production applications. Each project demonstrates expertise in architecture design, clean code practices, and delivering real business value through technology.